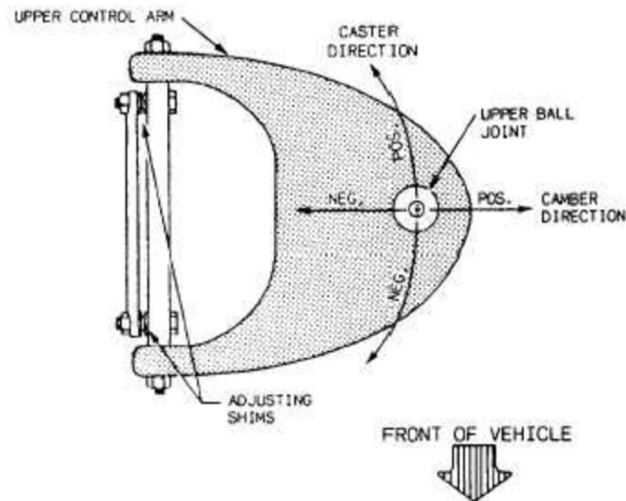
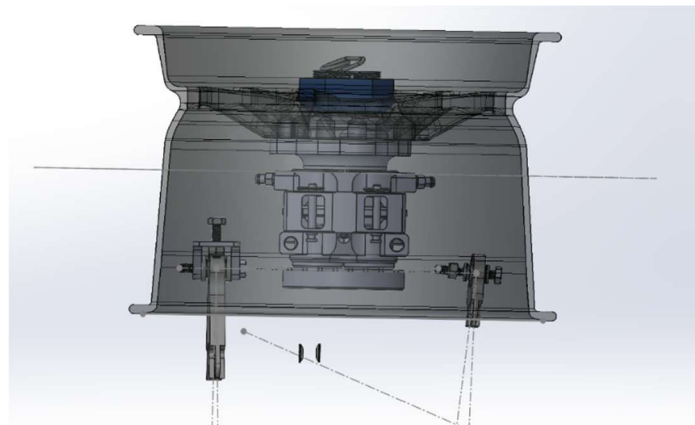


Shims Report

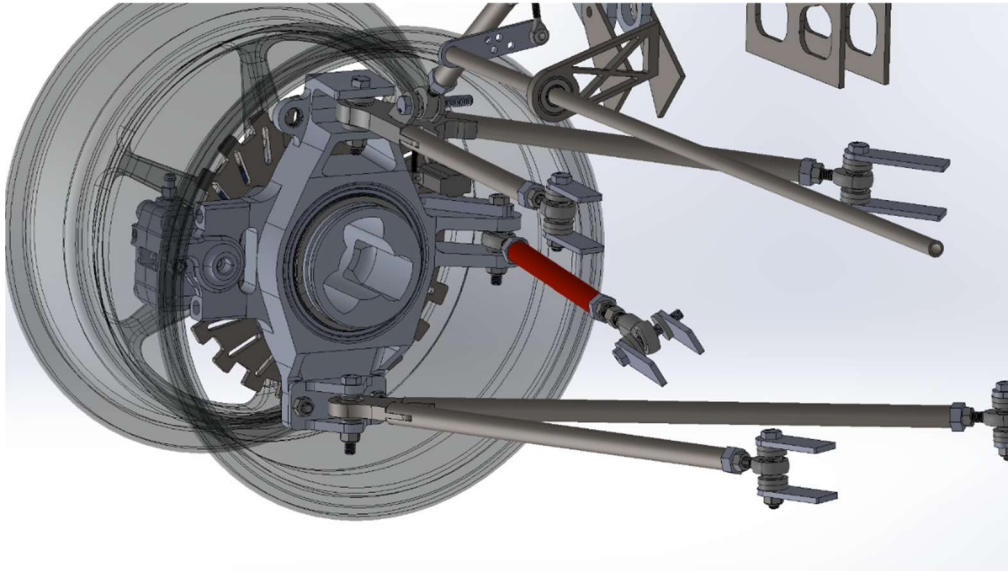
Two questions are to be answered here: why is Formula Duke placing camber shims on the lower arm, and why do most teams place their camber shims on the top control arm mount?



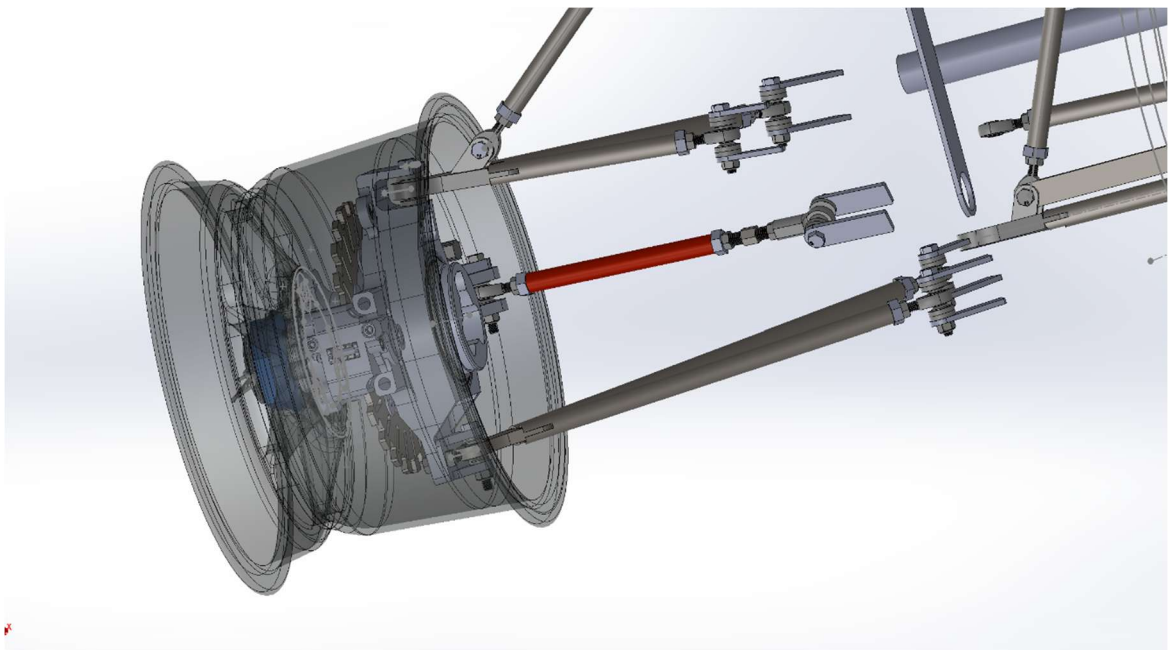
The mechanism is straightforward: shimming a mount pushes that ball joint outboard, away from the chassis. Adding shims at the upper mount moves the top of the upright outboard for more positive camber, so a team running negative camber removes shims to get there. Adding shims at the lower mount instead pushes the bottom outboard, tilting the top of the wheel inward for more negative camber, which is how our setup behaves. Because race cars run negative camber, the two layouts differ in how many shims sit at the corner at the working setting: an upper-mount car carries few shims at its negative baseline, a lower-mount car carries more. That difference is only a few grams, however, so weight is a minor consideration at best. The substantive trade-offs are serviceability and load path, addressed below.



Looking at our 2026 setup, we see further evidence in support of placing shims on the bottom. Based on the CAD, the lower upright pickup appears to have been selected primarily for accessibility and ease of shim-based adjustment. The lower clevis naturally accommodates spacer packs and provides excellent tool access, whereas the upper pickup is integrated into a much denser region of the upright.



From a kinematic standpoint either pickup can set the same static camber; the differences lie in load path and in secondary effects on scrub radius and kingpin inclination. For our upright the lower clevis is markedly easier to service, so our choice was driven primarily by serviceability and manufacturability.





The same static camber can be reached from either mount, but the two are not mechanically identical: they differ in load path and in their effect on other alignment parameters. Most teams nonetheless prefer the upper mount. Why?

As we note on this vehicle, other teams prefer having shims on the upper mount.

Two things make the upper mount attractive. First, load path: in a typical double wishbone the upper ball joint carries less load than the lower one, which also reacts braking and most of the suspension load, so a shimmed, slotted-bolt joint sits more cleanly out of the primary load path when placed up top. Second, geometry isolation: the lower joint sits near the ground, so moving it laterally shifts the steering-axis ground intercept and changes scrub radius and kingpin inclination noticeably, whereas moving the upper joint pivots the axis about the lower one and adjusts camber with far less disturbance to scrub radius. The shim mass is negligible against the wheel, tire, and upright, so teams trade it for clean, repeatable, trackside adjustment, with slotted packs behind the upper carrier letting the pickup move laterally without touching the lower joint or the wishbone lengths.

Online forums have an illuminating answer to this query as well –

“Our upper ball-joint carriers bolt to the uprights, and you shim between them to add positive camber (static setting is pretty negative, and you adjust to kick it back out).

The thing you want to think through is how you will add or take away shims. Most camber shims have slotted bolt holes so that you don't have to take the bolts completely off to change shims, just loosen them, pull the components slightly apart, slide the shims in or out, and then re-torque.”

Both choices are therefore well founded: Formula Duke gains real serviceability and packaging benefits from the lower clevis, while teams on the upper mount gain a cleaner load path and camber adjustment that barely disturbs scrub radius, with the small shim mass being immaterial either way.